Service Bulletin

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QSK45 and QSK60 Cylinder Block Machining Procedure to Upgrade to Straight Split Connecting Rods	

QSK45 and QSK60 Cylinder Block Machining Procedure to Upgrade to Straight Split Connecting Rods

Purpose

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This Service Bulletin outlines the block machining requirements to upgrade old QSK45 and QSK60 cylinder blocks to accept the straight split connecting rod. Cummins Inc. customers are **not** required to upgrade to the straight split connecting rod at rebuild. Angle split connecting rods will continue to be available for both rebuild and in-field repairs. The straight split connecting rod and related hardware was developed to provide improved durability in all applications. New cylinder block castings accept the new connecting rod and piston cooling nozzles without this machining. Straight split connecting rods can **only** be used with new-style piston cooling nozzles and cylinder blocks which contain the new casting changes or cylinder blocks that have been upgraded per this procedure. Cylinder blocks built prior to approximately March 2008 require machining to allow for the new hardware.

Five machining operations (See Figure 1) are required per bay (pair of cylinders) to provide clearance for the new connecting rod which is larger at the big end than the previous angle split (serrated) connecting rod. New piston cooling nozzles, piston cooling nozzle bolts, and installation process are also required. The party that authorizes the machine shop to perform the block machining is accountable for ensuring consistent quality in the machining operations.

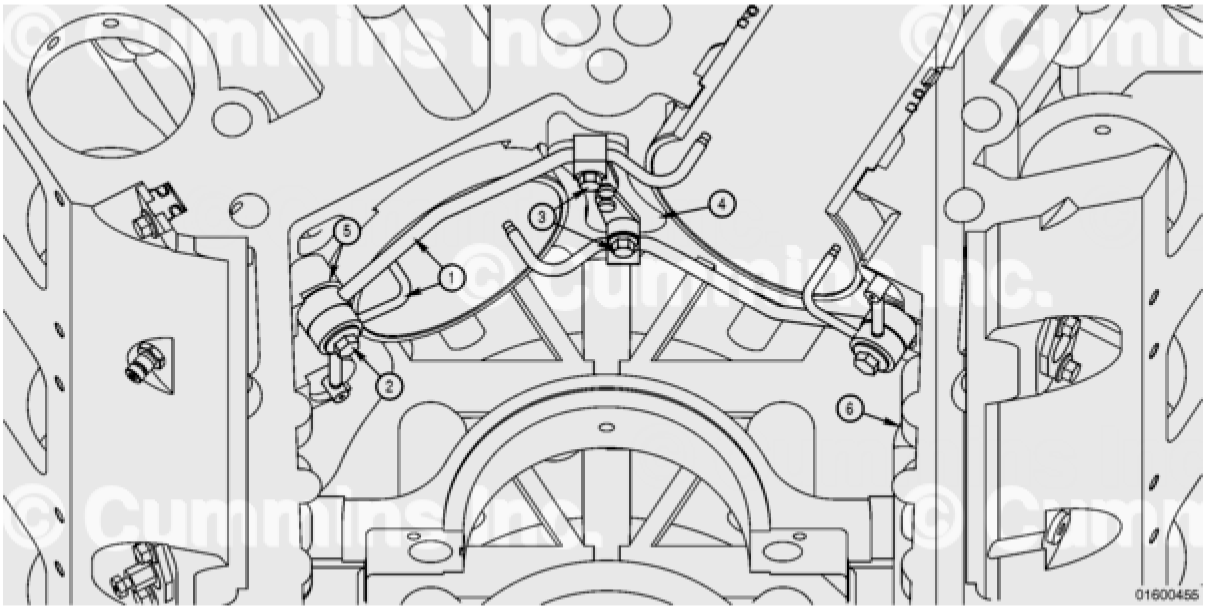


Figure 1: Overview of Machining and New Requirements for Upgrade (1) New Piston Cooling Nozzles, (2) New Piston Cooling Nozzle Banjo Bolt, (3) New Piston Cooling Nozzle Center Bolt, (4) Machined Upper Piston Cooling Nozzle Mounting Pad, (5) Machined Lower Piston Cooling Nozzle Mounting Pad - two locations, (6) Machined Side Wall Block - two locations.

Upper Piston Cooling Nozzle Mounting Pad (See Figure 2)

- Remove and discard the existing dowels prior to machining.
- Machine the upper piston cooling nozzle mounting pad as described later in this bulletin.
- Do **not** increase the depth of dowel holes or threads. Internal oil passages can be pierced and depth will affect the interference of the press fit dowel pins.
- Install new dowel pins.

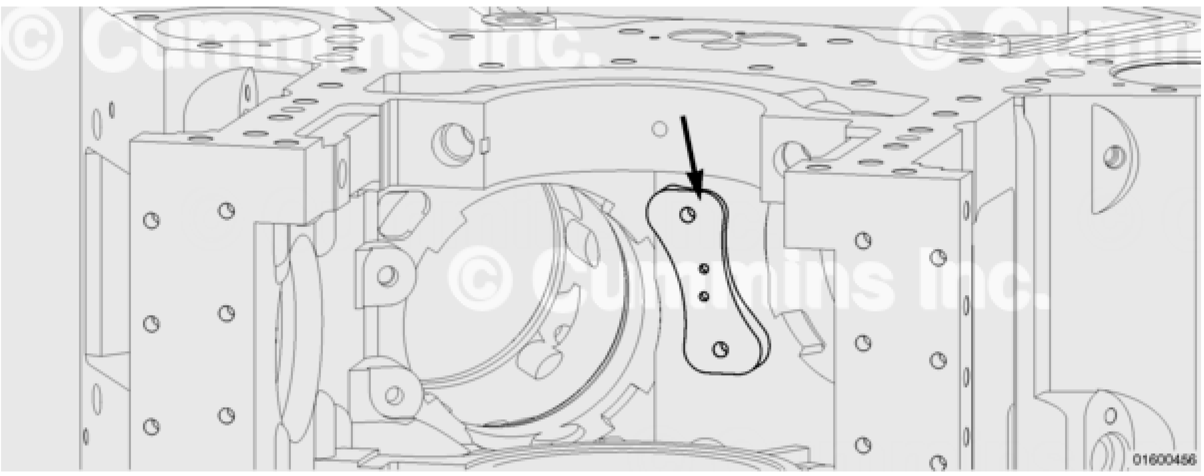


Figure 2: Upper Piston Cooling Nozzle Mounting Pad

Lower Piston Cooling Nozzle Mounting Pad (See Figure 3)

- **Only** machine the piston cooling nozzle mounting pad, **not** the piston cooling nozzle locating pad.

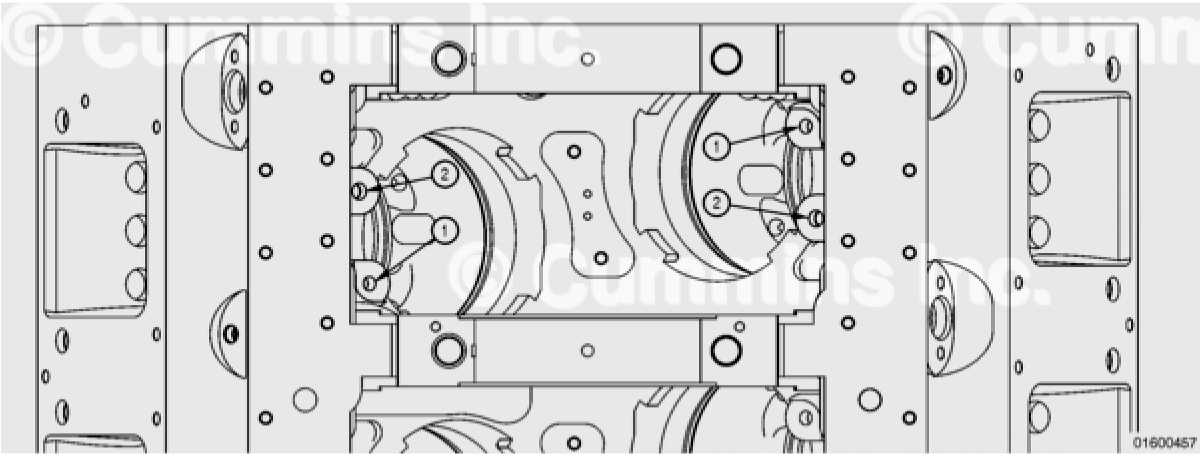


Figure 3: Lower Piston Cooling Nozzle Mounting Pad (1) Locating Pad Primary Datum [Piston Cooling Nozzle Locating Pad], (2) Banjo Bolt Mounting Pad [Piston Cooling Nozzle Mounting Pad].

Sidewall (See Figure 4 and 5)

⚠ CAUTION ⚠

Only remove material as specified by this service bulletin. The removal of excess material can increase material stress in other areas of the block.

- Remove a minimal amount of material from the sidewall (1). Variations in the cylinder block casting can result in no material being removed.
- A 2.38 mm [0.093 in] minimum tool radius is required; up to 10 mm [0.39] maximum tool radius is acceptable as long as the surface finish is maintained.
- Datums (2) used for machining the sidewalls are shown in the Figure 4.
- Sidewall machining can expose hand hole cover capscrew holes. Use a thread sealant or silicone sealant on the hand hole cover capscrews during final assembly. This will reduce the possibility of an oil leak from the hole.

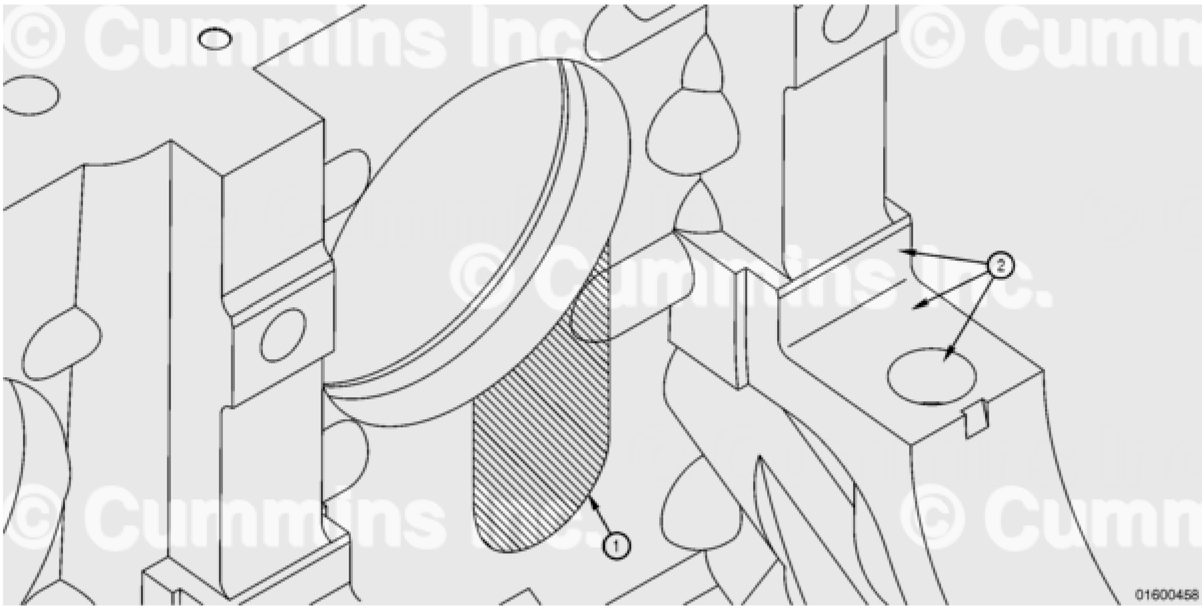


Figure 4: Sidewall Machining (1) Sidewall, (2) Datums used for machining.

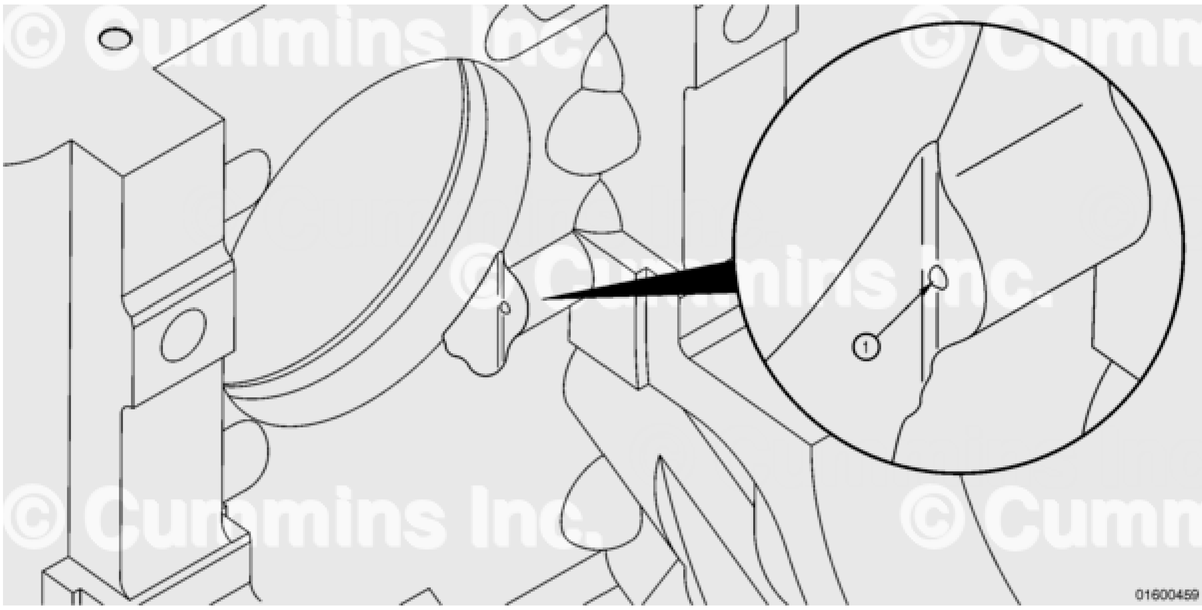


Figure 5: Sidewall Machining (1) Capscrew hole exposed by sidewall machining; use a silicone sealant on the hand hole cover capscrews during final assembly to reduce the possibility of an oil leak.

Marking Cylinder Block for Completed Machining (See Figure 6)

- Mark the cylinder block with an “A1” to show the machining has been completed.
- The marking **must** be located on the cylinder block pan rail, approximately 50 mm [2 in] in front of the engine serial number (ESN).

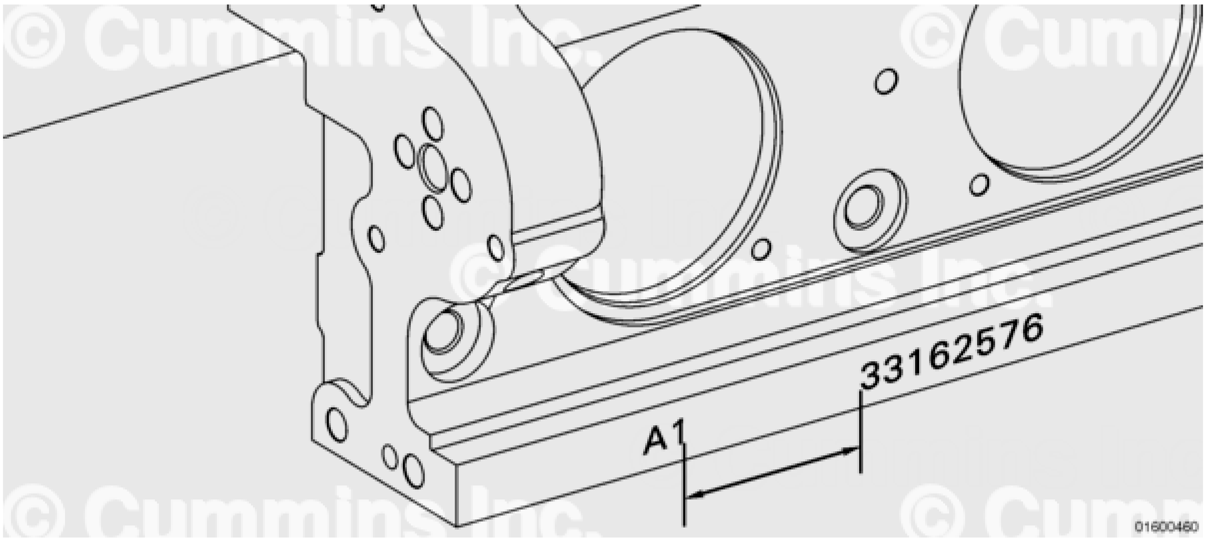


Figure 6: Completed cylinder block, marked with “A1”

Requirements that apply to all machining operations when **not** specified on prints:

- All machining print measurements are in millimeters
- Final Surface Finish: 3.2 μm
- All Tolerances: +/- 0.25 mm [0.0098 in]
- All radii unless specified: 0.8 mm [0.031 in] (NOTE: Side wall cut radii is a minimum of 2.38 mm [0.093 in])
- The party that authorizes the machine shop to perform the block machining is accountable for ensuring consistent quality in the machining operations.

Upper Piston Cooling Nozzle Mounting Pad (See Figure 7 and 8)

- Figure 7 shows the location of Datum A (1) and the upper piston cooling nozzle mounting pad (2).
- Figure 8 shows an engineering drawing of machining details.

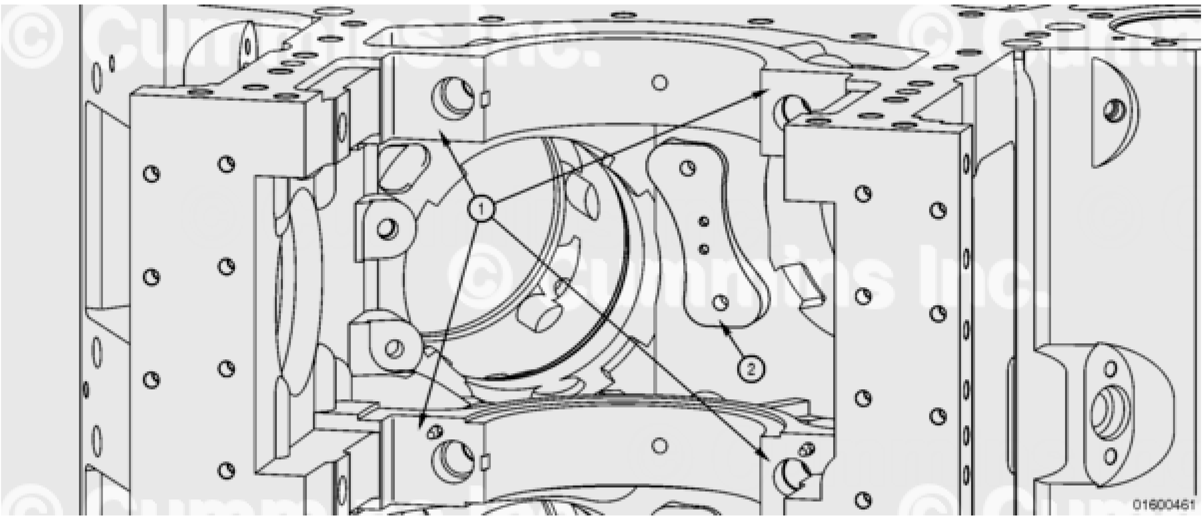


Figure 7: Upper Piston Cooling Nozzle Reference Points (1) Datum A, (2) Mounting Pad

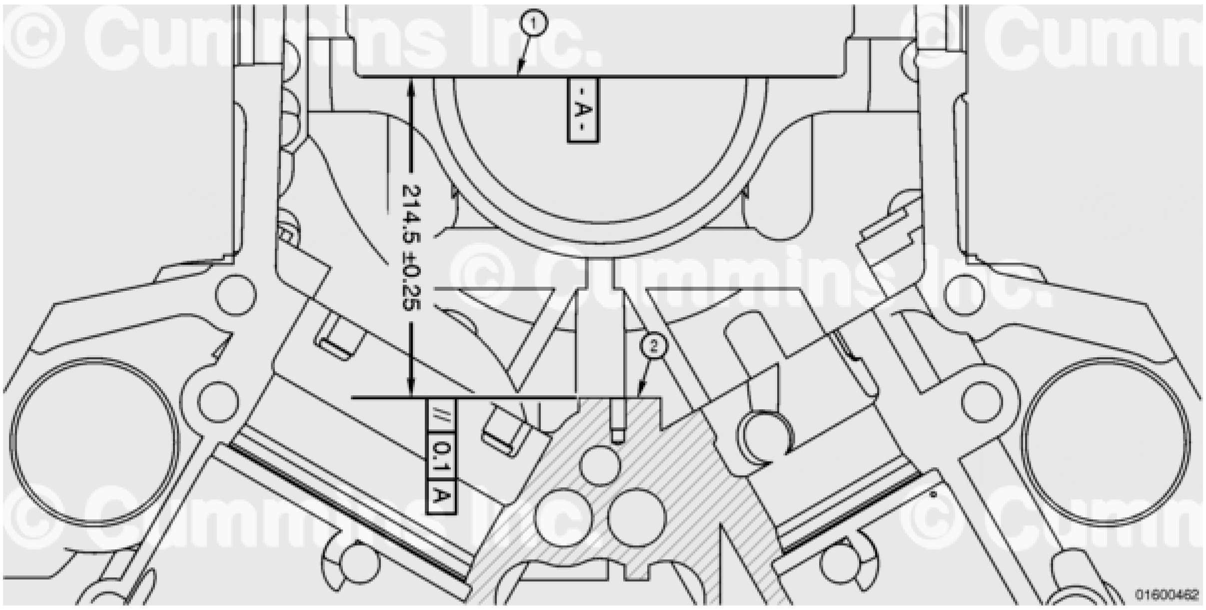


Figure 8: Upper Piston Cooling Nozzle Mounting Pad (1) Datum A, (2) Mounting Pad

Lower Piston Cooling Nozzle (See Figure 9, 10, and 11)

- Figure 9 shows the location of the primary datum reference surface (1) or locating pad and the mounting pad (2).
- Figure 10 shows an engineering drawing of machining details.
- NOTE: Surfaces (1) and (2) in Figure 9 and Figure 10 are parallel to the cylinder head mounting surface (head deck).
- A tool radius of 0.8 mm [0.031 in] **must** be used.
- Figure 11 shows how to locate an alternate datum for the machining process.

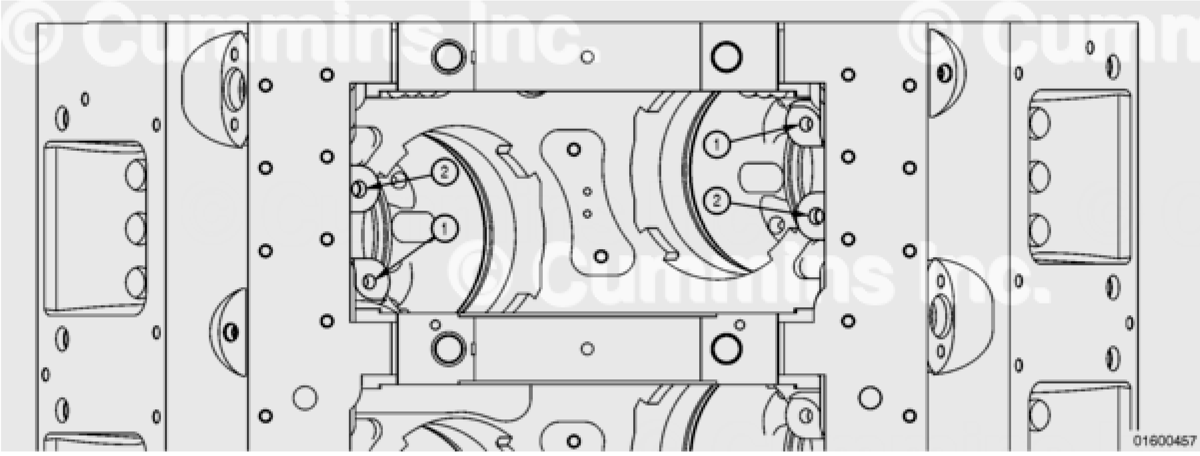


Figure 9: Lower Piston Cooling Nozzle Reference Point (1) Primary Datum Reference Surface (locating pad), (2) Mounting Pad.

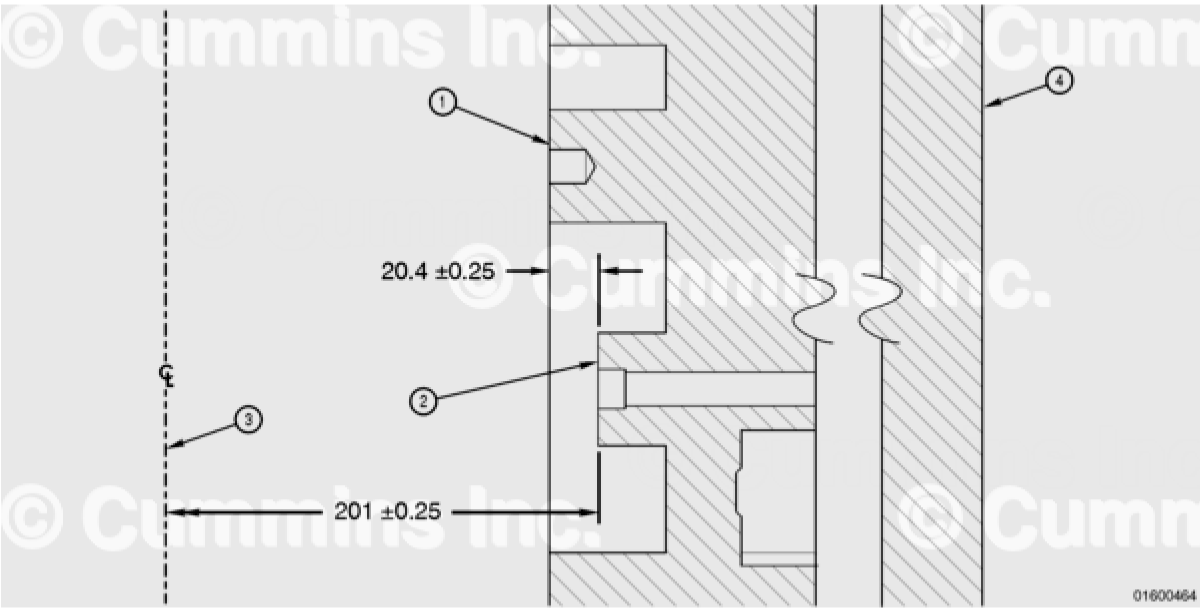


Figure 10: Lower Piston Cooling Nozzle (Oil Feed) (1) Primary Reference Datum (locating pad), (2) Surface to be Machined (mounting pad), (3) Alternate Datum (crankshaft centerline), (4) Cylinder Head Deck (parallel to locating pad and mounting pad).

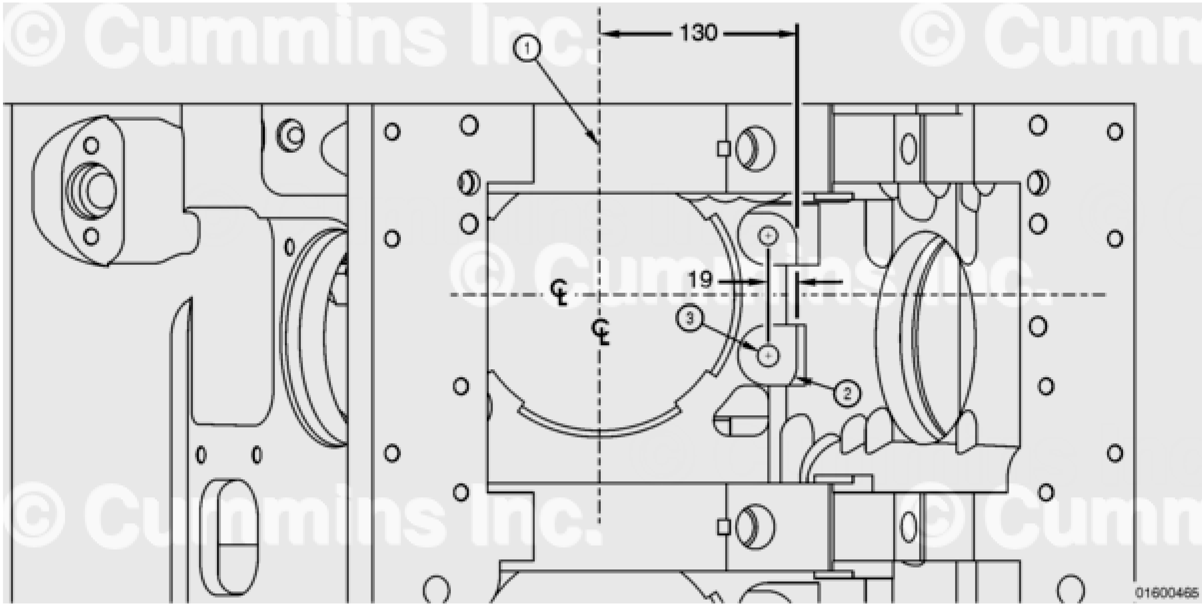


Figure 11: Lower Piston Cooling Nozzle Alternate Datum (1) Alternate Datum, (2) Corner can be removed, (3) M14 x 2 - 6H Thread.

Side Wall Machining (See Figure 12 and 13)

- Figure 12 and Figure 13 show engineering drawings of machining details. Be sure to use a flat surface from which to set the machining depth.

Note : The cutter tool must feature a 2.4 mm [0.094 in] tool radius; up to 10 mm [0.39] maximum tool radius is acceptable as long as the surface finish is maintained.

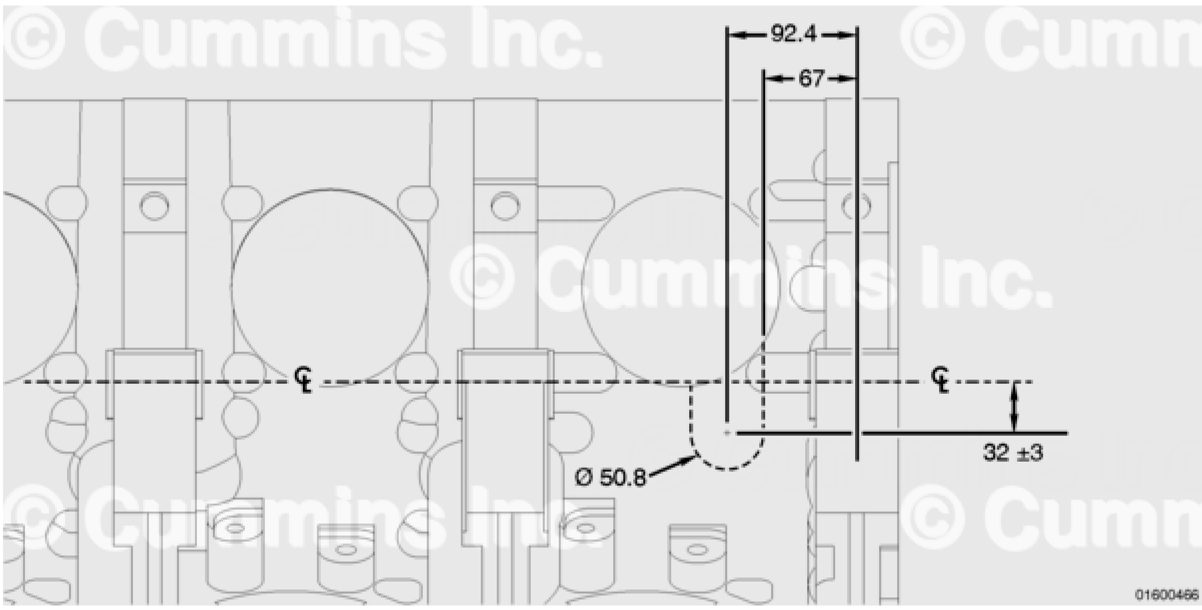


Figure 12: Side Wall Machining Location

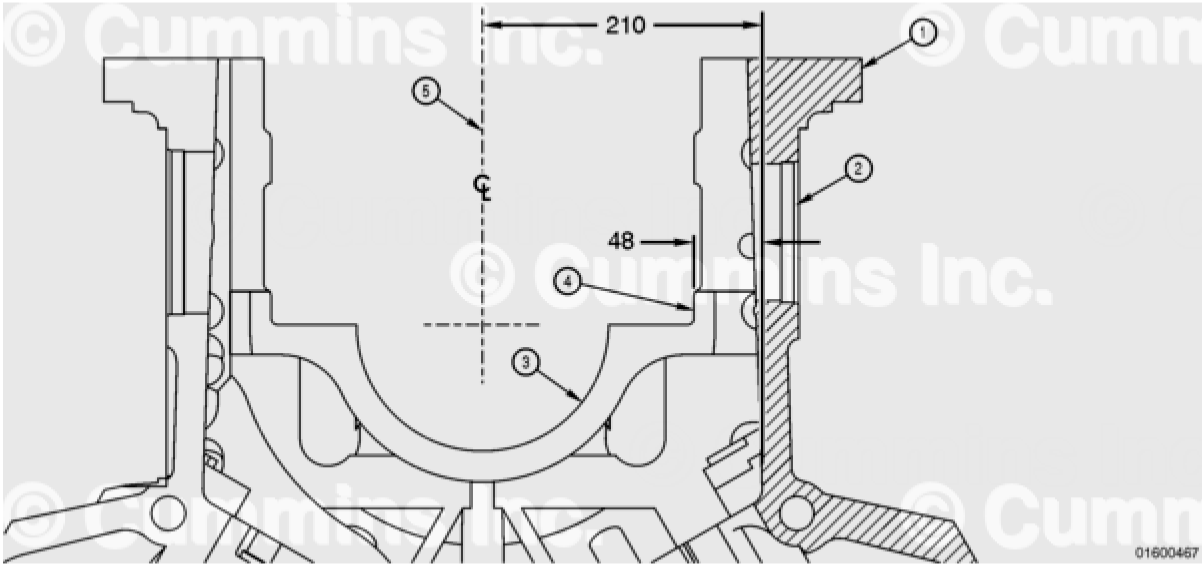
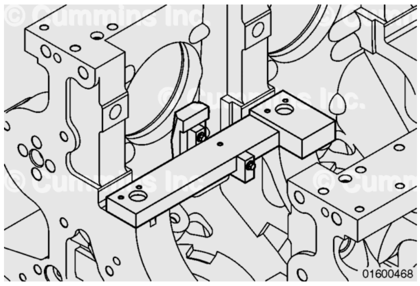


Figure 13: Side Wall Machining Depth (1) Pan Rail, (2) Hand Hole, (3) Main Bearing Saddle, (4) Flat Surface from which to set the Machining Depth, (5) Alternate Datum (crankshaft centerline).

Inspection Gauges

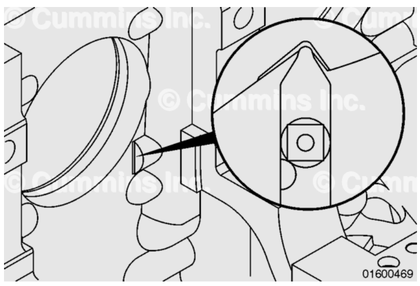


Block skirt machining minimum clearance gauge, Part Number 4918968, can be used to inspect for sufficient connecting rod clearance. It has been designed to include tolerances of crankshaft, connecting rod, and bearing clearances. If the tool will fit the cylinder block as shown without binding, then adequate clearance of the connecting rod is verified.

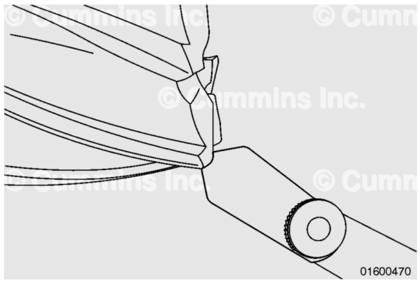
Note : This tool inspects for minimum required clearance. There may be excess clearance up to the dimensions allowed on the machining prints.

Additional gauges can be manufactured or purchased locally and used to ensure proper machining has been completed. The following illustrations are examples of locally manufactured or purchased gauge tools that can be used to check machining quality and consistency.

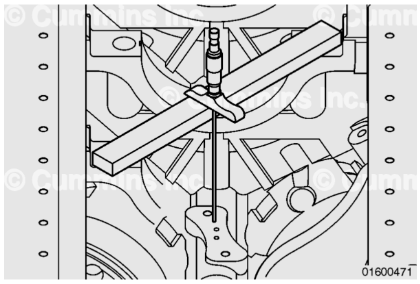
Internal Radius Gauge



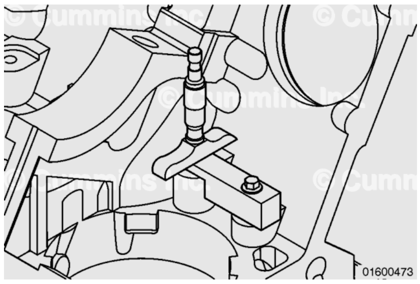
External Radius Gauge



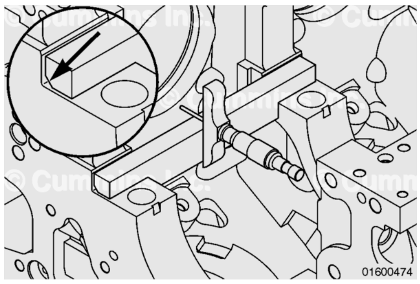
Upper Piston Cooling Nozzle Depth Gauge



Lower Piston Cooling Nozzle Depth Gauge



Side Wall Depth Gauge



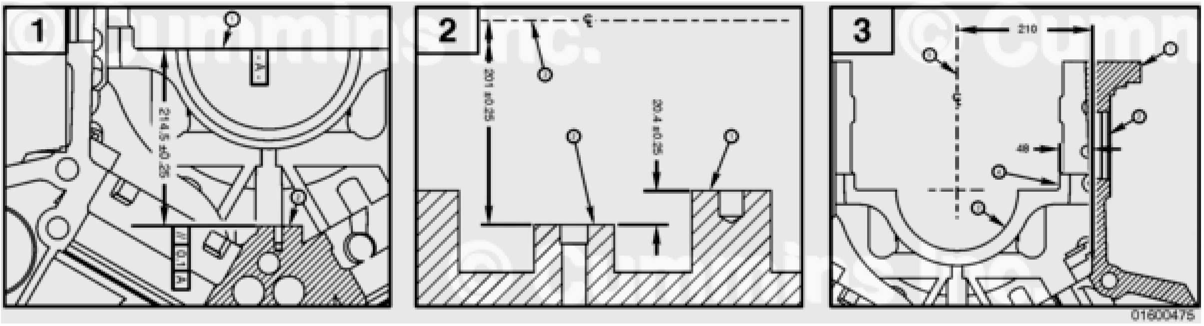
- Service/Parts Topic 08T1-31 New Procedure, Piston and Cylinder Liner
- Service/Parts Topic 08T1-33 New Cylinder Block and Piston Cooling Nozzle
- Service/Parts Topic 08T1-35 Revision, Multiple Sections, Piston Cooling Nozzle Procedure
- Service/Parts Topic 08T1-36 Revision, Multiple Steps, Connecting Rod Procedure
- Service/Parts Topic 08T1-40 New Straight Split Connecting Rods

A data sheet must be completed and retained to document the conformance of each block to specifications. A sample block machining data sheet is attached at the end of this bulletin.

Data Sheet for Updated QSK45 and QSK60 Cylinder Blocks for Straight Rods	
Job: ____________________	Date: ____________________

Data Sheet for Updated QSK45 and QSK60 Cylinder Blocks for Straight Rods

ESN: ____________________	Block S/N: ____________________



1. Upper Piston Cooling Nozzle Machining

Specification: 214.25 to 214.75 mm [8.435 to 8.454 in]

	CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8
A								
B								

1. Upper Piston Cooling Nozzle Machining
NOTE: Measurements must be taken at 2 separate locations (A and B) to check flatness.

Surface Finish Passed? YES / NO

2. Lower Piston Cooling Nozzle Machining
Specification: 20.15 to 20.65 mm [0.793 to 0.812 in]

CYL L1	CYL L2	CYL L3	CYL L4	CYL L5	CYL L6	CYL L7	CYL L8
CYL R1	CYL R2	CYL R3	CYL R4	CYL R5	CYL R6	CYL R7	CYL R8

Surface Finish Passed? YES / NO

3. Side Wall Machining							
Specification: 47.75 to 48.25 mm [1.879 to 1.899 in]							
CYL L1	CYL L2	CYL L3	CYL L4	CYL L5	CYL L6	CYL L7	CYL L8
CYL R1	CYL R2	CYL R3	CYL R4	CYL R5	CYL R6	CYL R7	CYL R8

Surface Finish Passed? YES / NO

Cylinder Block Marked with “A1”? YES / NO

Cylinder Block Cleaned? YES / NO